Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/Georgia/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/Henry/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Static · 36 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Static · 30 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/Nat/Static · 37 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/StephenV2/Static · 35 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/Mansh/Static · 36 trials
Total training paths	10 · 349 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/Josh/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats → feature dim 1408
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=558 · test=70

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Stacking (SVM+RF+KNN+LogReg)	0.9964	0.0107	1.0000
SVM (RBF)	0.9964	0.0107	0.9857
SVM (Linear)	0.9964	0.0107	0.9857
Logistic Regression	0.9946	0.0114	0.9857
Voting Soft (SVM+RF+KNN+LogReg)	0.9964	0.0107	0.9714
Random Forest	0.9946	0.0114	0.9571
KNN	0.9499	0.0327	0.9429
LDA	0.9355	0.0518	0.3429

Per-class hold-out F1

Class	Stacking	SVM (RBF)	SVM (Linear)	Logistic Regression	Voting (soft)	Random Forest	KNN	LDA	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	0.952	0.957	0.353	11
Double_Nothing	1.000	0.941	0.941	0.941	0.875	0.875	0.875	0.333	9
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	0.947	0.222	10
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	0.952	0.273	10
Double_Phone	1.000	1.000	0.952	1.000	0.952	0.952	0.900	0.182	10
Double_Pistol	1.000	1.000	1.000	1.000	1.000	0.952	1.000	0.429	10
Double_Spiderman	1.000	0.952	1.000	0.952	0.952	0.952	0.952	0.462	10
macro avg	**1.000**	**0.985**	**0.985**	**0.985**	**0.969**	**0.955**	**0.941**	**0.322**	**70**

External test (NewTestData)

Algorithm	External test acc

Random Forest	0.8857
Logistic Regression	0.8571
Voting Soft (SVM+RF+KNN+LogReg)	0.8571
Stacking (SVM+RF+KNN+LogReg)	0.8286
SVM (Linear)	0.7714
SVM (RBF)	0.7429
KNN	0.6000
LDA	0.1714

Per-class external accuracy

Class	Random Forest	Logistic Regression	Voting (soft)	Stacking	SVM (Linear)	SVM (RBF)	KNN	LDA	Support
Double_Closed_Fist	1.000	0.800	0.800	0.600	0.800	0.600	0.400	0.000	5
Double_Nothing	0.200	0.200	0.200	0.200	0.200	0.200	0.000	0.200	5
Double_Okay	1.000	1.000	1.000	1.000	0.800	0.800	0.200	0.000	5
Double_Open_Palm	1.000	1.000	1.000	1.000	0.800	0.800	1.000	0.000	5
Double_Phone	1.000	1.000	1.000	1.000	0.800	0.800	0.600	0.200	5
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.200	5
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.600	5